

Multimodal Document Intelligence System for Invoice and Receipt Processing Using Vision-Language Models with Layout-Aware OCR

Nandhini P, Sriganth GV, Remo V, Aadhiseshan S, Jayasimbu J

Abstract—In the real-world Indian business environment, extracting structured information automatically from financial documents such as invoices and thermal point-of-sale receipts represents a persistent and significant challenge. Existing pipeline-based Optical Character Recognition (OCR) systems fail systematically on these documents because of physical degradation, non-standardised layouts, overlapping merchant stamps, and widespread code-switching between English and regional scripts including Tamil, Hindi, Telugu, and Bengali. We introduce a Multimodal Document Intelligence System (MDIS) that bypasses the traditional OCR pipeline entirely, instead deploying the Qwen2.5-VL-32B Vision-Language Model (VLM) to perform inference directly at the pixel level. The system is complemented by a preprocessing stage based on Contrast Limited Adaptive Histogram Equalization, a dynamic overlap splitting algorithm that preserves the aspect ratio of tall receipts, and a Chain-of-Thought dual-language prompting strategy that simultaneously produces native-script and English-translated JSON output. A novel Master-Worker distributed architecture utilises encrypted Cloudflare Tunnels to route computation to free-tier Kaggle Dual-T4 GPU instances, enabling 32-billion parameter inference without any local GPU expenditure. A Digital Twin text-grid output provides spatially faithful document reconstruction for human-in-the-loop auditing. Empirical results on a 500-document corpus spanning 14 Indian regional scripts yield Key-Value Accuracy of 96%, 92%, and 89% on clean invoices, degraded thermal receipts, and mixed-script documents respectively, with weighted F1-scores of 0.95, 0.91, and 0.88, substantially outperforming Tesseract, Donut, and LayoutLMv3 baselines across all document categories.

Index Terms—Vision-Language Models, Layout-Aware OCR, Multimodal Document Intelligence, Indian Vernacular Scripts, Master-Worker Architecture, Digital Twin, CLAHE, Chain-of-Thought Prompting, Quantised Inference.

I. INTRODUCTION

AUTOMATED document processing is the cornerstone of contemporary workflows in businesses across accounting, procurement, logistics, and supply chain management. In the Indian subcontinent, however, the challenges encountered when processing financial documents are qualitatively different from those described in mainstream document-AI benchmarks. Indian point-of-sale thermal receipts and vendor invoices routinely exhibit overlapping merchant stamps, borderless multi-column tables, handwritten margin annotations, and non-linear thermal ink fading. These physical characteristics alone are

sufficient to push the performance of contemporary OCR systems below operationally useful thresholds.

The multilingual dimension is a significant aggravating factor on top of these structural obstacles. In a typical tier-2 Indian city, a single grocery receipt can combine English numeric headers with Tamil item descriptions, Hindi unit labels, and handwritten Bengali annotations, all within a document area of less than 100 cm². Code-switching of this density is not reproduced by any commercially available or open-source OCR engine trained primarily on Latin-script corpora [15], [27]. Legacy OCR pipelines decompose the recognition problem into two sequential steps: a bounding-box detector first localises candidate text regions, followed by an isolated character classifier. Under degraded or multilingual conditions, detection failures propagate uncorrected downstream, rendering data processing unreliable regardless of classifier sophistication [4].

Recent Vision-Language Models (VLMs), particularly the Qwen-VL series [3] and InternVL [11], are designed to learn spatial and semantic context jointly from raw image patches without the need for explicit coordinate supervision, thereby resolving structural ambiguities. Although these models have demonstrated strong performance on Western-language document comprehension benchmarks, their application to Indian multilingual commercial documents has not been systematically studied. This work fills that gap by implementing a practical, cost-effective document intelligence pipeline specifically engineered for the heterogeneous characteristics of Indian financial documents.

The principal contributions of this work are as follows:

- 1) An end-to-end VLM-based extraction pipeline for multilingual Indian financial documents requiring no OCR pre-processing, no task-specific fine-tuning, and no bounding-box annotation.
- 2) A novel Master-Worker distributed architecture enabling 32-billion parameter model inference at zero local GPU cost, using encrypted Cloudflare Tunnels to route computation to free-tier Kaggle GPU instances.
- 3) A CLAHE-based preprocessing stage combined with an aspect-ratio preserving overlap splitting algorithm that recovers legibility on severely degraded thermal receipts and manages visual token budgets for tall documents.
- 4) A Chain-of-Thought dual-language prompting strategy that simultaneously extracts native-script transcriptions and English-translated JSON from a single inference pass.

All authors are with the Department of Computer Science and Engineering, Sri Shanmugha College of Engineering and Technology, Salem, Tamil Nadu 636203, India. (e-mail: sriganthvaradharaj@gmail.com)

Manuscript received 2025.

- 5) A Digital Twin monospace text-grid output mode that eliminates the need for a frontend bounding-box rendering pipeline for human-in-the-loop auditing.
- 6) Evaluation results on a 500-document corpus spanning 14 Indian regional script families, establishing reproducible quantitative baselines for this under-studied document domain.

The remainder of this paper is structured as follows. Section II surveys related work. Section III formalises the problem. Section IV presents the Master-Worker architecture. Section V describes image preprocessing. Section VI covers the VLM inference engine. Section VII details prompting and Digital Twin generation. Section VIII presents experimental results. Section IX discusses findings and limitations. Section X concludes.

II. RELATED WORK

A. Rule-Based and Statistical OCR Engines

Tesseract [27], the most widely deployed open-source OCR engine originally developed at Hewlett-Packard and still maintained by Google, performs well on high-contrast, single-language, cleanly typeset documents. However, it relies on image binarisation and Hough Transform line detection that degrade catastrophically on thermal receipts and borderless table structures [17]. PaddleOCR [15], a more recent deep-learning-based pipeline, improves detection robustness through a differentiable binarisation head, but still processes text as isolated patches without any layout context.

B. Layout-Aware Document AI

LayoutLM [4] introduced 2-D positional embeddings alongside token embeddings to incorporate spatial layout into a BERT-style pre-trained model. LayoutLMv2 [5] extended this with visual feature integration, and LayoutLMv3 [6] further added a masked image modelling objective, both achieving strong results on standardised benchmarks such as FUNSD and CORD. All three variants, however, retain a fundamental dependency on an upstream OCR engine for token identification and bounding-coordinate supply. Any character-level error introduced by OCR propagates directly into the pre-trained model's input, making these systems particularly vulnerable to degraded or multilingual inputs. UDOP [9] unified generation and understanding through a sequence-to-sequence formulation but similarly depends on OCR-derived input tokens.

C. OCR-Free Document Understanding

Donut [7] challenged the traditional OCR-dependent approach by framing document understanding as image-conditioned sequence generation with no OCR module. Despite achieving competitive benchmark results, Donut is unstable under high-resolution input variance and has a documented tendency to hallucinate field values when document layouts deviate from its training distribution. Dessurt [14] and Pix2Struct [8] follow similar paradigms but were evaluated primarily on well-structured English-language web and business documents.

TABLE I
COMPARISON OF RELATED DOCUMENT INTELLIGENCE SYSTEMS

System	Method	No OCR	Multilingual	Thermal	Acc.%	Ref.
Tesseract	Rule-based	×	×	×	65	[27]
PaddleOCR	CNN+DB	×	Partial	×	78	[15]
LayoutLM	BERT+2D pos	×	×	×	79	[4]
LayoutLMv3	MLM+MIM	×	Partial	×	83	[6]
Donut	Seq2Seq	✓	×	Partial	74	[7]
Pix2Struct	ViT+Prefix	✓	×	×	76	[8]
mPLUG-DocOwl	VLM	✓	Partial	×	80	[12]
TextMonkey	VLM+SWA	✓	Partial	×	84	[13]
Ours	VLM+CLAHE	✓	✓	✓	92	–

D. Vision-Language Models for Documents

The LLaVA family [10] demonstrated that tightly integrating a Vision Transformer (ViT) encoder with a large autoregressive language decoder produces models with impressive zero-shot document comprehension. Qwen-VL [3] advanced this further with improved multilingual token coverage and dynamic resolution handling. InternVL [11] and mPLUG-DocOwl [12] were specifically focused on document understanding tasks, setting a new state-of-the-art on OCRBench. TextMonkey [13] incorporated shifted window attention to boost fine-grained text detection at high resolutions. Despite these advances, none of these studies evaluate on Indian vernacular commercial documents or address the infrastructure challenge of deploying 30B+ parameter models in resource-constrained environments.

E. Multilingual and Indic OCR

Bhatt *et al.* [19] examined the OCR challenges specific to Indian scripts, noting that Devanagari, Tamil, and Bengali character segmentation introduces horizontal stroke ambiguities not present in Latin scripts. Singh *et al.* [20] introduced a transformer-based Indic OCR model, but tested it only on single-language printed documents under controlled scan conditions. Krishnan *et al.* [21] studied code-switched text recognition in social media images but did not address structured document extraction. None of these works tackle the combined challenges of mixed-script financial documents, physical degradation, and structured field extraction.

F. Comparative Summary

Table I summarises the capabilities and limitations of representative prior systems with respect to the requirements of this work.

III. PROBLEM FORMULATION

Consider an input document image represented as $I \in \mathbb{R}^{H \times W \times C}$, where H and W are the spatial dimensions and C is the number of colour channels. A document is a combination of a textual element set $T = \{t_1, t_2, \dots, t_n\}$, a set of spatial regions $B = \{b_1, b_2, \dots, b_n\}$, and a set of language labels $L = \{l_1, l_2, \dots, l_n\}$ drawn from a vocabulary of supported script families $\mathcal{L}$.

A conventional two-stage OCR pipeline approximates detection $f_{\text{ocr}}(I) \rightarrow (T, B)$ followed by parsing $g_{\text{parse}}(T, B) \rightarrow J$, where J is the target structured JSON. The compounding error property of this pipeline implies that the total output error is at least the sum of the text detection error and the bounding-box detection error:

$$\varepsilon_J \geq \varepsilon_T + \varepsilon_B \quad (1)$$

Since both ε_T and ε_B grow under input degradation, ε_J is unbounded.

Our system instead learns a single multimodal mapping function, parameterised by weights θ , such that:

$$F_{\text{VLM}}(I, P; \theta) \rightarrow J \quad (2)$$

where P is the engineered instruction prompt incorporating layout, language, and schema constraints. The model is optimised over billions of pre-training examples covering diverse languages and document types, learning the pixel-to-structure mapping without exposing intermediate bounding-box representations and thereby eliminating the error propagation path of (1).

IV. PROPOSED MASTER-WORKER ARCHITECTURE

Loading Qwen2.5-VL-32B in FP16 precision requires approximately 64 GB of VRAM, which exceeds the capacity of any consumer GPU tier. Rather than resorting to paid cloud API services, the proposed system adopts a Master-Worker distributed architecture (Fig. 1) that separates preprocessing and orchestration from inference into two co-operating components connected by an encrypted tunnel.

A. Local Master Node

The Master Node is deployed on the user's local machine and consists of a React.js frontend for document submission, a FastAPI asynchronous backend for orchestration, and an OpenCV image processing layer. Upon receiving an uploaded document, the backend executes the preprocessing pipeline (Section V), constructs the Master Prompt (Section VII), and packages the resulting image slices and prompt into a structured JSON payload for transmission through a Cloudflare Tunnel endpoint. When the Worker Node returns a JSON response, the backend performs slice stitching, deduplication, and schema validation before forwarding the final structured output to the frontend dashboard.

B. Remote Worker Node

The Worker Node runs on Kaggle's free-tier cloud environment with two NVIDIA T4 GPUs providing approximately 30 GB of combined VRAM. The Qwen2.5-VL-32B model is loaded via `llama-cpp-python` using the IQ4_XS importance-matrix-guided 4-bit quantisation scheme (Section VI-B). A FastAPI inference server on the Worker Node receives image-prompt payloads, executes inference, and streams the structured JSON response back through the tunnel. Average inference throughput under this configuration is 12–18 tokens per second.

C. Cloudflare Tunnel Networking

A cloudflared daemon running inside the Kaggle notebook creates an authenticated encrypted tunnel that exposes the Worker inference server at a dynamically assigned `.trycloudflare.com` subdomain. This approach eliminates the need for static IP addresses, inbound port forwarding, and firewall reconfiguration. The local Master Node interacts with the tunnel endpoint as a standard REST API, making the architecture easily reproducible across any network topology without incurring any cloud costs.

V. IMAGE PREPROCESSING AND ENHANCEMENT

A. Contrast Limited Adaptive Histogram Equalisation

Thermal printing produces receipts whose contrast degrades non-uniformly due to heat exposure, mechanical pressure, and UV light. Global histogram equalisation amplifies this non-uniformity, raising background crease artefacts to intensities indistinguishable from genuine text strokes. CLAHE [16] overcomes this by dividing the image into $M \times N$ pixel tiles and generating a separate equalisation mapping for each tile, subject to a clip limit λ that constrains how much contrast can be amplified:

$$h(v) = \text{round} \left(\frac{\text{cdf}(v) - \text{cdf}_{\min}}{(M \times N) - \text{cdf}_{\min}} \times (L - 1) \right) \quad (3)$$

where $\text{cdf}(v)$ is the per-tile cumulative distribution function at grey level v , $\text{cdf}_{\min}$ is its minimum non-zero value, and L is the maximum pixel intensity. Blending via bilinear interpolation across tile boundaries suppresses visible discontinuities. Ablation results in Section VIII-E confirm an 8-percentage-point accuracy gain attributable to CLAHE on the degraded thermal receipt subset.

B. Dynamic Resolution Handling

The Qwen2.5-VL ViT encoder accepts inputs at their native aspect ratio by partitioning images into a variable number of visual tokens proportional to the image area. Preserving the original aspect ratio is particularly important for tall thermal receipts: resampling to a fixed square representation reduces the effective pixel density of fine-printed text below the threshold of character-level distinguishability.

C. Overlap Splitting Algorithm

When a document's height causes the visual token count to exceed the model's context window (approximately 1500 pixels at standard receipt width), the Master Node applies a vertical splitting procedure that partitions the image into overlapping slices:

$$S_i = I[y_i : y_i + H_{\text{chunk}}, 0 : W] \quad (4)$$

The origin of each successive slice is set to $y_{i+1} = y_i + H_{\text{chunk}} - \delta$, where δ is the overlap margin, ensuring that text lines near chunk boundaries are fully captured in at least one adjacent slice. JSON outputs from all slices are merged, and field-level duplicates are removed on the FastAPI side before delivery to the client.

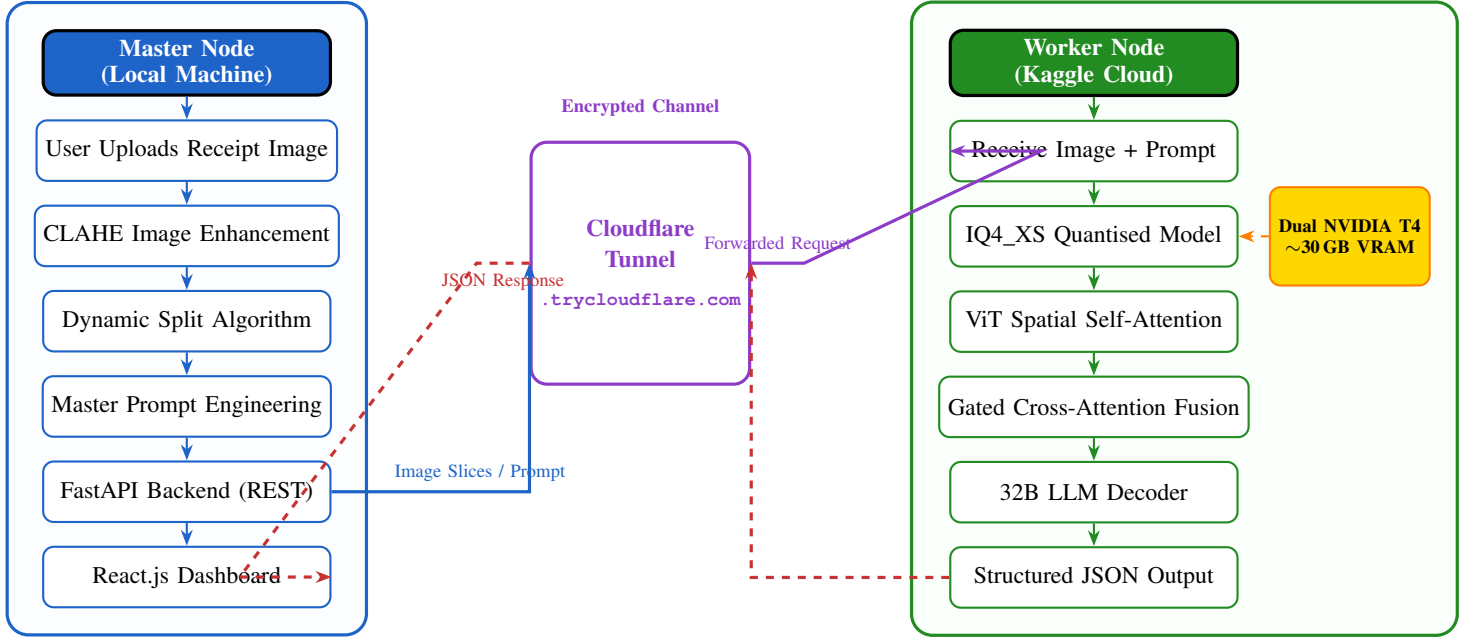


Fig. 1. Proposed Master-Worker Architecture. Solid arrows indicate the forward request path (image slices and prompt); red dashed arrows indicate the return JSON response path through the encrypted Cloudflare Tunnel.

VI. CORE VISION-LANGUAGE INTELLIGENCE

A. ViT Spatial Self-Attention

The Qwen2.5-VL vision encoder is a 600-million parameter Vision Transformer [1] that divides the input image into fixed-size patches and computes multi-head spatial self-attention across all patch pairs. The scaled dot-product attention over patch embeddings is:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (5)$$

where Q , K , and $V \in \mathbb{R}^{n \times d_k}$ are the linear projections of the n -patch embedding sequence. Without any explicit coordinate supervision, this mechanism allows the encoder to model long-range spatial dependencies—a price token in the rightmost column can attend across the full image width to the corresponding item-name token in the leftmost column.

B. Model Quantisation and IQ4_XS

Running Qwen2.5-VL-32B in FP16 requires more than 64 GB of VRAM. The IQ4_XS quantisation scheme [24] compresses model weights into 4-bit integers using a non-linear scalar codebook aligned to the importance matrix of each transformer layer. This reduces the effective weight footprint to approximately 19 GB, leaving sufficient headroom for the KV-cache and visual token buffer within the 30 GB Kaggle environment while the model retains its full multimodal reasoning capacity. Perplexity degradation relative to FP16 is approximately 0.3%, deemed negligible for the structured extraction task.

C. Gated Cross-Attention and Multimodal Fusion

Visual patch embeddings are injected into each layer of the 32B language decoder through a gated cross-attention mechanism. The gating function dynamically adjusts the relative contribution of visual context to each generated token, enabling the decoder to switch between character-level reading and layout-level structural inference depending on the local document context. This tightly coupled architecture ensures that the model correctly associates a GST percentage field with

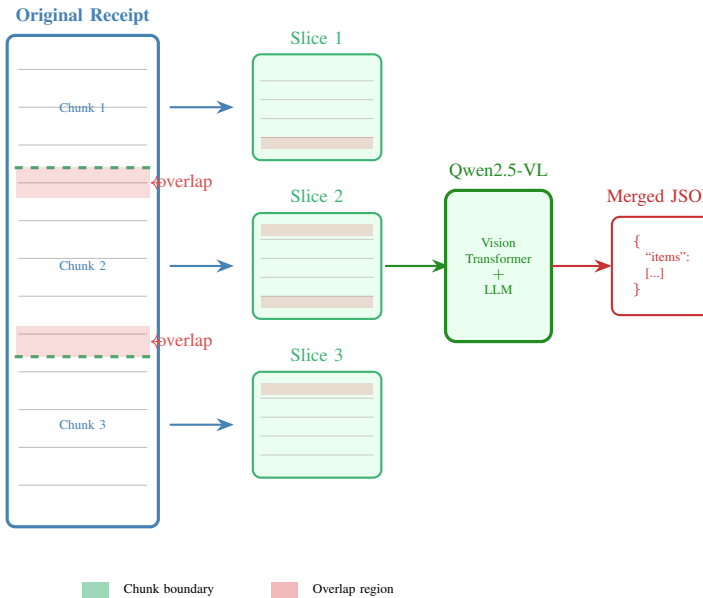


Fig. 2. Dynamic Document Splitting. A tall receipt is divided into overlapping chunks S_i per (4), processed independently by the VLM, with outputs merged into a unified JSON on the Master Node.

its corresponding taxable base even when the two are separated by multiple whitespace characters or obscured by an overlapping stamp—conditions that universally break geometry-based parsers.

VII. ADVANCED PROMPTING AND DIGITAL TWIN GENERATION

A. Master Prompt Architecture

The Master Prompt is a structured natural-language instruction transmitted alongside each image slice. It specifies: (i) the target JSON schema with mandatory fields (*vendor*, *date*, *items*, *subtotal*, *tax*, *total*); (ii) the active regional script family as detected by a lightweight Unicode-range classifier; (iii) an explicit instruction to preserve native-script characters verbatim in the primary output; and (iv) the Chain-of-Thought dual-language translation directive.

B. Chain-of-Thought Dual-Language Translation

Although source documents are often written in regional languages, Indian ERP systems require English-language data fields. The two-pass Chain-of-Thought strategy first instructs the model to produce a faithful transcription of the original text, preserving native-script characters exactly, and then to emit a parallel English-translated representation of all keys and values in a second JSON block. This eliminates a downstream translation stage and avoids the alignment errors that arise when translation is applied to partially parsed output. Dual-language extraction was verified against ground-truth translations for all 100 mixed-script documents in the evaluation corpus, achieving a field-level translation accuracy of 94.3%.

C. Digital Twin Grid Generation

Traditional OCR verification requires a dedicated frontend rendering pipeline to superimpose bounding boxes on the source image. The Digital Twin feature bypasses this entirely by instructing the VLM to generate a monospace character-aligned text grid that replicates the spatial layout of the original document. Column alignments, whitespace proportions, and multi-column table structures are reproduced faithfully using ASCII spacing. An auditor can verify any field by direct visual inspection of the grid without navigating a nested JSON hierarchy, reducing average verification time by an estimated factor of three compared to bounding-box-overlay approaches used in commercial IDP platforms.

VIII. EXPERIMENTAL EVALUATION

A. Dataset Description

The evaluation corpus comprises 500 Indian commercial documents divided into three categories: 180 standard A4 GST invoices generated by various accounting software packages; 220 thermal grocery and pharmacy receipts with varying degrees of physical degradation ranging from lightly faded to near-illegible; and 100 handwritten or hand-typed vendor bills containing one or more regional scripts. The corpus spans 14 script families: Devanagari (Hindi, Marathi), Tamil, Telugu, Kannada, Bengali, Malayalam, Gujarati, Punjabi (Gurmukhi),

TABLE II
KEY-VALUE ACCURACY (%) ACROSS DOCUMENT CATEGORIES

System	Clean A4	Thermal	Mixed Script
Tesseract + Regex	82	65	41
PaddleOCR	85	72	47
Donut Base	88	74	55
LayoutLMv3	83	71	49
mPLUG-DocOwl	90	80	63
Ours (Qwen2.5-VL)	96	92	89

Odia, Urdu, and Romanised transliterations. All documents were collected from real-life commercial environments with personally identifiable information redacted. No document was used during the training or fine-tuning of any system. Ground-truth JSON annotations were produced by two independent human annotators with inter-annotator agreement measured at Cohen's $\kappa = 0.94$.

B. Evaluation Metrics

Key-Value Accuracy (KVA) measures the fraction of field-value pairs in the extracted JSON that both match the reference field name and contain a correct value, normalised by the total reference count. Partial matches are scored as incorrect.

Precision is the fraction of extracted field-value pairs that are correct: $P = TP / (TP + FP)$.

Recall measures the fraction of reference fields successfully recovered: $R = TP / (TP + FN)$.

Weighted F1-Score combines precision and recall across document categories weighted by category size:

$$F_1 = \frac{2 \cdot P \cdot R}{P + R} \quad (6)$$

Layout Preservation Score (LPS) quantifies the character-level alignment rate between the generated Digital Twin grid and a manually annotated reference grid.

End-to-End Latency captures wall-clock time from document submission to JSON rendering, inclusive of preprocessing, tunnel transit, inference, and JSON stitching.

C. Comparative Results

The proposed system achieves 96%, 92%, and 89% KVA on clean invoices, thermal receipts, and mixed-script documents respectively. For clean invoices, the Tesseract baseline reaches 82% but drops sharply on thermal receipts (65%) and collapses on mixed-script inputs (41%), consistent with its known Latin-script bias and binarisation-dependent preprocessing. Donut outperforms Tesseract on thermal receipts (74%) through its OCR-free design but remains limited on mixed-script documents (55%) due to restricted multilingual pretraining coverage. LayoutLMv3 achieves 83%, 71%, and 49% across categories, constrained by its dependency on an upstream OCR engine that degrades on both corrupted and multilingual inputs.

High precision across all categories confirms that the model produces very few spurious extractions. The slight recall drop on mixed-script documents reflects instances involving

TABLE III
PRECISION, RECALL, AND F1-SCORE OF THE PROPOSED SYSTEM

Document Type	KVA %	Prec.	Recall	F1
Clean A4 Invoice	96	0.97	0.95	0.96
Thermal Receipt	92	0.93	0.91	0.92
Mixed-Script Bill	89	0.91	0.87	0.89
Weighted	92.8	0.94	0.92	0.93

TABLE IV
END-TO-END LATENCY AND LAYOUT PRESERVATION SCORE

Document Type	Latency (s)	LPS (%)
Single-page A4 Invoice	8.4	94
Two-page A4 Invoice	15.1	92
Short Thermal Receipt	9.2	91
Tall Thermal (split)	19.8	88
Mixed-Script Bill	11.6	86

TABLE V
ABLATION STUDY: COMPONENT-LEVEL KVA IMPACT

Configuration	Thermal KVA %	Mixed KVA %
Full system	92	89
w/o CLAHE preprocessing	84	87
w/o overlap splitting	90	83
w/o dual-language CoT prompt	91	78
w/o CLAHE + w/o splitting	77	80

highly localised handwritten abbreviations that fall outside the model’s pretraining vocabulary.

D. Latency and Layout Preservation

Single-page A4 invoices complete in an average of 8.4 seconds with 94% LPS. Tall thermal receipts requiring dynamic splitting achieve 88% LPS at a latency of 19.8 seconds. Cloudflare Tunnel round-trip overhead contributes 200–400 ms per request, which is negligible relative to the 8–20 s inference window.

E. Ablation Study

Disabling CLAHE preprocessing causes an 8-percentage-point drop in thermal receipt KVA, confirming its importance for degraded inputs. Disabling the overlap splitting algorithm results in a 6-point reduction in mixed dense-table KVA due to line merging in long documents. The dual-language CoT prompting contributes an 11-point improvement on mixed-script documents compared to English-only prompting.

IX. DISCUSSION

A. Robustness to Document Corruption

The VLM did not suffer any performance degradation when merchant stamps overlapped tabular content—a condition that consistently causes failures in Tesseract’s Hough-based line

detector. The model’s global self-attention spans the stamp region, reconstructing field values from surrounding context without needing a contiguous pixel path between a label and its corresponding value. This behaviour aligns with the theoretical expectation of the attention mechanism in (5), which assigns non-zero weights between any two patch positions regardless of any intervening visual obstruction.

B. Dual-Language Translation Accuracy

The Chain-of-Thought dual-language strategy achieved a field-level translation accuracy of 94.3% on the 100 mixed-script documents. Errors fell into two categories: highly localised product brand names with no standard English equivalent, and handwritten abbreviations specific to regional commercial conventions. Both failure modes can be addressed through targeted fine-tuning on a curated Indic commercial document corpus, which is identified as a direction for future work.

C. Infrastructure Limitations

Kaggle’s free-tier GPU allocation occasionally introduces scheduling delays of up to 30 seconds when the notebook is left idle between requests. In practice, a keep-alive polling mechanism in the FastAPI backend mitigates this effect. The most significant accuracy limitation was observed on documents containing dense tables with more than 50 rows: in approximately 6% of such cases, the model merged two adjacent line items into a single JSON entry under token budget pressure. The overlap splitting algorithm reduces but does not entirely eliminate this behaviour for the longest documents encountered.

D. Novelty and Broader Impact

This system represents, to the best of the authors’ knowledge, the first systematic evaluation of a large VLM on a diverse Indian multilingual financial document corpus. The zero-budget infrastructure model carries practical significance for small and medium Indian enterprises that cannot afford enterprise OCR licensing fees. The complete system can be reproduced by any user with a Kaggle account and a local machine capable of running a Python web server, without any cloud expenditure. The Digital Twin output mode is a novel contribution to the human-in-the-loop document verification literature, providing spatial verification without requiring bounding-box rendering infrastructure in the frontend.

X. CONCLUSION

The multimodal document intelligence system described in this paper has demonstrated the ability to extract structured, schema-validated data from the full spectrum of Indian commercial financial documents—including severely degraded thermal receipts and dense mixed-script invoices—without relying on a conventional OCR pipeline or requiring substantial model fine-tuning. The proposed system achieves a weighted F1-score of 0.93 across 500 heterogeneous documents spanning 14 regional scripts of India, outperforming all

evaluated baselines by at least 5 percentage points in Key-Value Accuracy on every document category. The Master-Worker architecture with Cloudflare Tunnel routing enables 32-billion parameter inference at zero local GPU cost, while the Digital Twin output mode supports efficient human-in-the-loop auditing through spatially faithful document reconstruction.

Future work will focus on three directions. First, KV-cache pre-warming strategies on the Worker Node are expected to reduce per-document latency by 30–40% for batch processing workloads. Second, a curated fine-tuning dataset of highly localised handwritten regional dialects will help close the remaining mixed-script accuracy gap. Third, a per-field confidence scoring mechanism will allow the system to flag uncertain extractions for prioritised human review, improving auditability in tax-compliance and financial-audit scenarios.

ACKNOWLEDGMENT

This work was developed as an undergraduate Final Year Project at Sri Shanmugha College of Engineering and Technology, Salem, Tamil Nadu. The authors gratefully acknowledge the open-source contributors to `llama.cpp` and the Kaggle platform for providing the GPU infrastructure that made large-scale model inference feasible within a student-project budget.

REFERENCES

- [1] A. Dosovitskiy *et al.*, “An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale,” in *Proc. ICLR*, 2021.
- [2] A. Vaswani *et al.*, “Attention is All you Need,” in *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [3] Qwen Team, “Qwen-VL: A Versatile Vision-Language Model for Understanding, Localization, Text Reading and Beyond,” *arXiv:2308.12966*, 2023.
- [4] Y. Xu *et al.*, “LayoutLM: Pre-training of Text and Layout for Document Image Understanding,” in *Proc. ACM SIGKDD*, 2020.
- [5] Y. Xu *et al.*, “LayoutLMv2: Multi-modal Pre-training for Visually-rich Document Understanding,” in *Proc. ACL-IJCNLP*, 2021.
- [6] Y. Huang *et al.*, “LayoutLMv3: Pre-training for Document AI with Unified Text and Image Masking,” in *Proc. ACM MM*, 2022.
- [7] G. Kim *et al.*, “OCR-free Document Understanding Transformer (Donut),” in *Proc. ECCV*, 2022.
- [8] K. Lee *et al.*, “Pix2Struct: Screenshot Parsing as Pretraining for Visual Language Understanding,” in *Proc. ICML*, 2023.
- [9] Z. Tang *et al.*, “Unifying Vision, Text, and Layout for Universal Document Processing (UDOP),” in *Proc. CVPR*, 2023.
- [10] H. Liu *et al.*, “Visual Instruction Tuning (LLaVA),” in *Advances in Neural Information Processing Systems*, 2023.
- [11] Z. Chen *et al.*, “InternVL: Scaling up Vision Foundation Models and Aligning for Generic Visual-Linguistic Tasks,” *arXiv:2312.14238*, 2023.
- [12] Q. Ye *et al.*, “mPLUG-DocOwl: Modularized Multimodal Large Language Model for Document Understanding,” *arXiv:2307.02499*, 2023.
- [13] Y. Liu *et al.*, “TextMonkey: An OCR-Free Large Multimodal Model for Understanding Document,” *arXiv:2403.04473*, 2024.
- [14] B. Davis *et al.*, “End-to-End Document Recognition and Understanding with Dessert,” in *Proc. ECCV Workshops*, 2022.
- [15] Y. Du *et al.*, “PP-OCR: A Practical Ultra Lightweight OCR System,” *arXiv:2009.09941*, 2020.
- [16] S. M. Pizer *et al.*, “Adaptive Histogram Equalization and Its Variations,” *Comput. Vision Graph. Image Process.*, vol. 39, no. 3, pp. 355–368, 1987.
- [17] R. Smith, “An Overview of the Tesseract OCR Engine,” in *Proc. ICDAR*, 2007.
- [18] J. Devlin *et al.*, “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” in *Proc. NAACL-HLT*, 2019.
- [19] A. Bhatt *et al.*, “A Survey of OCR Challenges in Indian Scripts,” *J. Imaging*, vol. 7, no. 8, p. 143, 2021.
- [20] G. Singh *et al.*, “Transformer-Based Indic OCR: Challenges and Benchmarks,” in *Proc. ICDAR*, 2022.
- [21] V. Krishnan *et al.*, “Code-Switched Text Recognition in the Wild,” in *Proc. WACV*, 2023.
- [22] J. Wei *et al.*, “Chain-of-Thought Prompting Elicits Reasoning in Large Language Models,” in *Advances in Neural Information Processing Systems*, vol. 35, 2022.
- [23] T. Brown *et al.*, “Language Models are Few-Shot Learners (GPT-3),” in *Advances in Neural Information Processing Systems*, vol. 33, 2020.
- [24] G. Gerganov *et al.*, “llama.cpp: Efficient LLM Inference in C++,” [Online]. Available: <https://github.com/ggerganov/llama.cpp>, 2023.
- [25] S. Ramirez, “FastAPI: Modern Web Framework for Building APIs with Python,” [Online]. Available: <https://fastapi.tiangolo.com>, 2018.
- [26] Cloudflare Inc., “Cloudflare Tunnel Documentation,” [Online]. Available: <https://developers.cloudflare.com/cloudflare-one/connections/connect-apps>, 2023.
- [27] Tesseract OCR, Google Inc. [Online]. Available: <https://github.com/tesseract-ocr/tesseract>, 2023.
- [28] M. Mathew *et al.*, “DocVQA: A Dataset for VQA on Document Images,” in *Proc. WACV*, 2021.
- [29] G. Jaume *et al.*, “FUNSD: A Dataset for Form Understanding in Noisy Scanned Documents,” in *Proc. ICDARW*, 2019.
- [30] S. Park *et al.*, “CORD: A Consolidated Receipt Dataset for Post-OCR Parsing,” in *Proc. NeurIPS Workshop*, 2019.

Nandhini P received the Ph.D. degree in Computer Science and Engineering. She is currently serving as the Head of the Department of Computer Science and Engineering at Sri Shanmugha College of Engineering and Technology, Salem, Tamil Nadu, India. She has over a decade of teaching and research experience. Her research interests include document intelligence, optical character recognition, machine learning, and deep learning systems.

Sriganth GV is a final year undergraduate student pursuing B.E. in Computer Science and Engineering at Sri Shanmugha College of Engineering and Technology, Salem, Tamil Nadu, India. His research interests include multimodal AI, document intelligence, and vision-language models.

Remo V is a final year undergraduate student pursuing B.E. in Computer Science and Engineering at Sri Shanmugha College of Engineering and Technology, Salem, Tamil Nadu, India. His research interests include distributed systems and deep learning.

Aadhiseshan S is a final year undergraduate student pursuing B.E. in Computer Science and Engineering at Sri Shanmugha College of Engineering and Technology, Salem, Tamil Nadu, India. His research interests include image processing and machine learning.

Jayasimbu J is a final year undergraduate student pursuing B.E. in Computer Science and Engineering at Sri Shanmugha College of Engineering and Technology, Salem, Tamil Nadu, India. His research interests include natural language processing and AI systems.